

AERO50001 Aerodynamics 2 Problem Set #2

Extra practice: Normal shocks, convergent-divergent nozzles

2.1 Isentropic nozzle

★☆☆ 15-18 mins

Helium, at $T_0 = 400$ K, enters a nozzle isentropically. At section 1, where $A_1 = 0.1$ m², a pitot static tube arrangement measures a stagnation pressure of 150 kPa and a static pressure of 123 kPa. For Helium, $\gamma = 1.66$, $R = 2077$ J/kg · K.

- (a) Estimate M_1 , the Mach number at section 1.
 - (b) Estimate T_1 , the temperature at section 1.
 - (c) Estimate the mass flow rate.
 - (d) Estimate A^* , the area at the throat.
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(a) At section 1, for isentropic flow:

$$\frac{p_0}{p_1} = \left(1 + \frac{\gamma-1}{2} M_1^2\right)^{\frac{\gamma}{\gamma-1}}$$

$$\therefore M_1 = 0.4988$$

(b) $T_0 = T_{01} = 400$ K

$$\frac{T_0}{T_1} = 1 + \frac{\gamma-1}{2} M_1^2 \quad \therefore T_1 = 369.65 \text{ K}$$

(c) $\dot{m} = \rho_1 u_1 A_1 = \rho_1 M_1 a_1 A_1 = \frac{p_1}{RT_1} M_1 \sqrt{\gamma RT_1} A_1$

$$= \frac{p_1 M_1 A_1 \sqrt{\gamma}}{\sqrt{RT_1}} = 9.021 \text{ kg/s}$$

$$(d) \left(\frac{A_1}{A^*}\right)^2 = \frac{1}{M_1^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_1^2\right) \right]^{\frac{\gamma+1}{\gamma-1}}$$

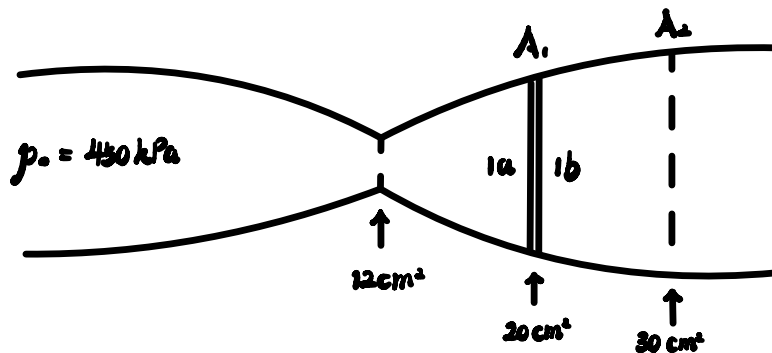
$$\therefore A^* = 0.0756 \text{ m}^2$$

2.2 Converging-diverging nozzle

★★☆ 15-20 mins

Air, supplied by a reservoir at 450 kPa flows through a converging-diverging nozzle whose throat area is 12 cm². A normal shock stands at section 1, where $A_1 = 20$ cm². Still further downstream is section 3, where $A_3 = 30$ cm².

- Compute the pressure just downstream of the shock at section 1.
- Estimate A_3^* , the area of the second throat required to choke the flow at section 3.
- Estimate the Mach number at section 3, M_3 .
- Estimate the pressure P_3 at section 3.



a) Just upstream of the normal shock, for isentropic process:

$$\frac{A_1}{A^*} = \frac{5}{3} \quad \therefore M_{1a} = 1.98 \text{ according to the figure}$$

$$\therefore \frac{p_0}{p_{1a}} = 7.5849 \quad \therefore p_{1a} = \frac{p_0}{7.5849}$$

$$\therefore \frac{p_{1b}}{p_{1a}} = 4.407 \quad \therefore p_{1b} = \frac{4.407}{7.5849} p_0 = 261.46 \text{ kPa}$$

b) $M_{1b} = 0.5808$ according to the figure

$$\therefore \left(\frac{A_1}{A_3^*} \right)^2 = \frac{1}{M_{1b}^2} \left[\frac{2}{\gamma+1} \left(1 + \frac{\gamma-1}{2} M_{1b}^2 \right) \right]^{\frac{\gamma+1}{\gamma-1}} = 1.4687$$

$$\therefore A_3^* = 16.5 \text{ cm}^2$$

$$c) \frac{A_2}{A_3^*} = 1.818$$

$\therefore M_3 = 0.34$ according to the figure

$$d) \frac{p_{01b}}{p_0} = 0.7302 \quad \therefore p_{01b} = 0.7302 p_0$$

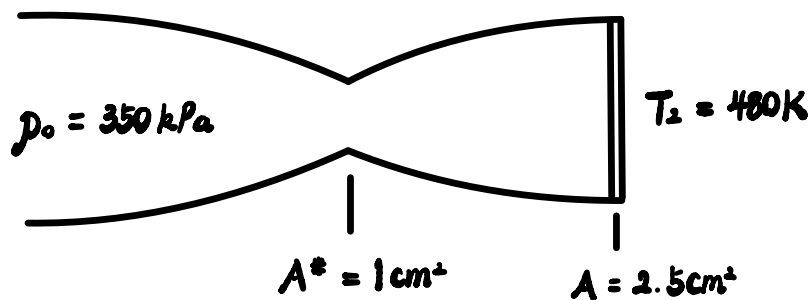
$$\frac{p_{01b}}{p_3} = 1.0833 \quad \therefore p_3 = \frac{0.7302}{1.0833} p_0 = 303.32 \text{ kPa}$$

2.3 Nozzle with shock

☆☆☆ 10-15 mins

A tank at 350 kPa feeds air ($\gamma = 1.4$) into a nozzle with a throat area of 1 cm^2 and an exit area of 2.5 cm^2 . A normal shock occurs at the exit of the nozzle, and just behind the normal shock the temperature is 480 K.

- (a) Calculate the temperature in the tank.
 - (b) Calculate the pressure behind the shock.
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$$(a) \frac{A}{A^*} = 2.5$$

$$\therefore M_1 = 2.45, \quad \frac{p_2}{p_1} = 6.836 \quad \text{and} \quad \frac{\rho_2}{\rho_1} = 3.273$$

$$\therefore \frac{T_2}{T_1} = \left(\frac{p_2}{p_1} \right) \left(\frac{\rho_1}{\rho_2} \right) = \frac{6.836}{3.273} \quad \therefore T_1 = 229.82 \text{ K}$$

$$\therefore \frac{T_0}{T_1} = 2.2005 \quad \therefore T_0 = 505.72 \text{ K}$$

$$(b) \frac{p_0}{p_1} = 15.806 \quad \therefore p_1 = 22.143 \text{ kPa}$$

$$\frac{p_2}{p_1} = 6.836 \quad \therefore p_2 = 151.37 \text{ kPa}$$